Ariyeh J. Weissman-Bennett

US Citizen | ariyeh18@gmail.com | (857)-285-8919 | www.linkedin.com/in/ari-weissman-bennett

PROFESSIONAL SUMMARY

Motivated fourth-year Mechanical Engineering student with experience in design, analysis, manufacturing, and engineering research. Hands-on experience with electromechanical systems, testing, and simulation through technical projects and laboratory work. Driven to apply technical knowledge and creativity to develop innovative engineering solutions.

EDUCATION

University of Massachusetts Amherst — **Amherst, Massachusetts**

B.S. in Mechanical Engineering — *Expected Graduation, May 2027*

- **GPA:** 3.71/4.00
- **Coursework:** Advanced Fluid Mechanics, Aerospace Structures, Thermodynamics, Design of Mechanical Components & Assemblies, Dynamic Systems, Heat Transfer, Electrical Engineering, Manufacturing Processes

SKILLS

Skills: SOLIDWORKS, MATLAB, Simulink, CAD, Ansys, Machine Learning, FDM 3D printing, Manual Machining, Arduino, Mechanical Design, DFMA, GD&T, Python, MS Office Suite

ENGINEERING EXPERIENCE

Undergraduate Research Intern — **Amherst, Massachusetts**

University of Massachusetts Amherst | THEAM Lab — *Aug 2026 – Present*

- Prepared pNIPAM hydrogel resin and fabricated custom microstructures using two-photon direct laser writing.
- Characterized temperature-dependent swelling and shrinking of pNIPAM hydrogels to investigate their potential as soft microactuators for controlling microfluidic flow.
- Investigated how microactuator geometry, and material properties influence response time and performance.
- Developing a biocompatible microfluidic fuel cell to power organ-on-a-chip devices using DLP 3D printing.
- Integrating PEGDA hydrogel barriers to separate oppositely charged fluids for voltage generation.

MathWorks Fault Detection for Electric Motors | Senior Capstone — **Amherst, Massachusetts**

Machine Learning & Code Lead — *Sep 2026 – Present*

- Developing and validating a Simulink electric motor model using Simscape Multibody to generate vibration data under healthy and various faulty operating conditions.
- Applying signal processing and feature extraction techniques to train machine-learning models in MATLAB.
- Integrating vibration sensors and microcontroller to collect motor signals for fault analysis and model validation.

Undergraduate Teacher Assistant | Strength of Materials & Additive Manufacturing — **Amherst, Massachusetts**

University of Massachusetts Amherst — *Jan 2026 – Present*

- Held office hours to assist students with core strength of materials subjects and mechanical design concepts.
- Supported students with CAD preparation, slicing, printer setup, and troubleshooting for FDM additive manufacturing.

Lightweight Wrench | Design Project — **Amherst, Massachusetts**

Team Member — *Sep 2025 – Dec 2025*

- Designed and optimized an aluminum wrench in SOLIDWORKS to meet geometric constraints and target weight.
- Performed hand stress analysis and Static Structural FEA in Ansys to validate safety factor prior to fabrication.
- Machined final part manually and validated performance through three-point bend and functional torque testing; presented results in final report and poster.

Mini Smart Car — **Amherst, Massachusetts**

Individual Contributor — *Dec 2025*

- Built a battery-powered smart car using an Arduino Uno, breadboarded circuitry, DC motors, and sensors.
- Programmed obstacle detection and avoidance for the car when objects were detected within 3 ft of the vehicle using integrated motor control and sensor feedback the used LED as status indicators and to signal system state.

HONORS, SCHOLARSHIPS & AWARDS

- Carl Christenson UMass Engineering Scholarship, Dean's List, Henry “Hank” J. Picard UMass Engineering Scholarship, Somerville Mathematics Fund Scholarship